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(54) PORTABLE CHAIR

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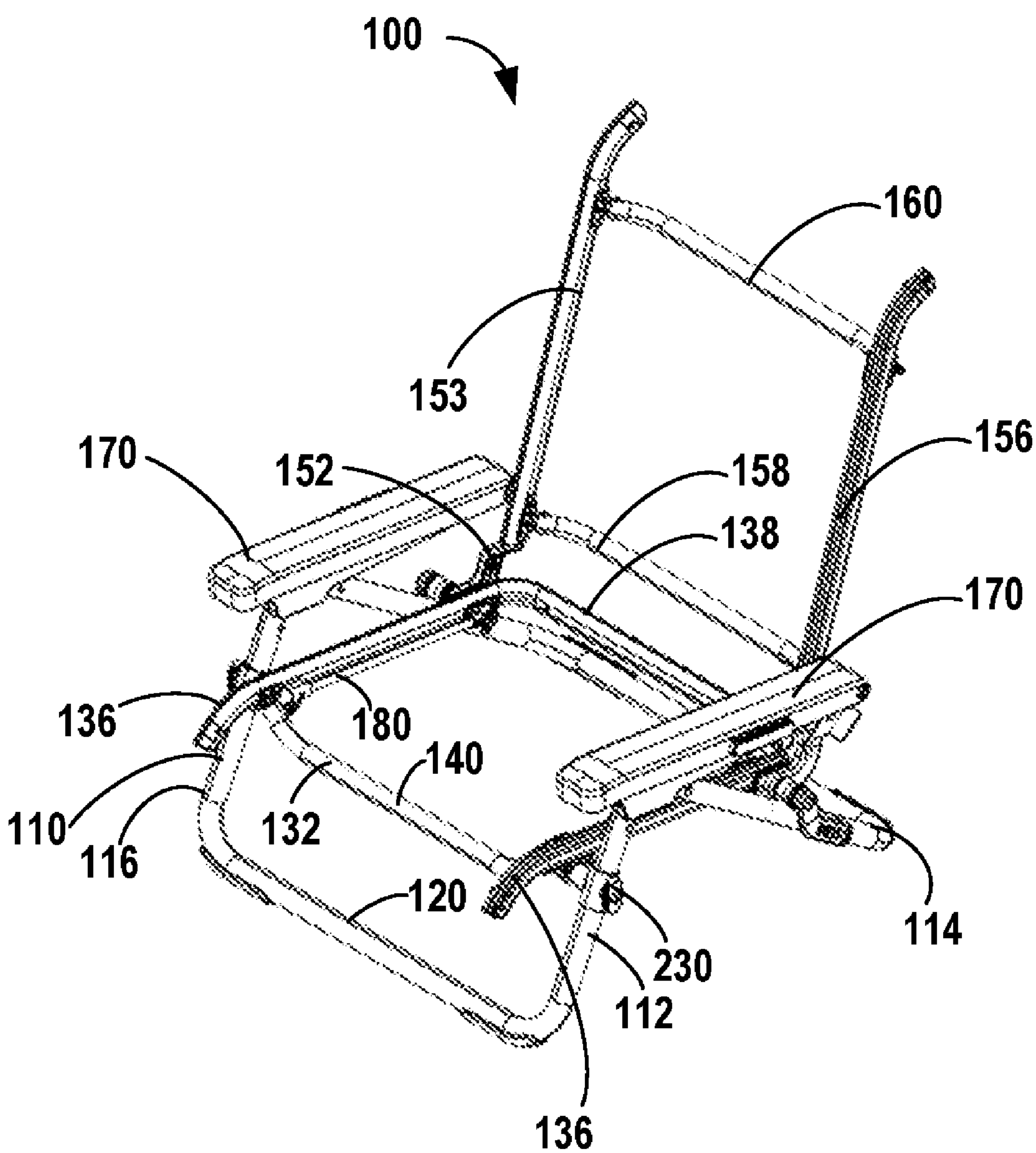
(57) **ABSTRACT**

A portable chair may include a first front leg, a first back leg pivotally connected to the first front leg, a second front leg, a second back leg pivotally connected to the second front leg, a seat support assembly and a seat assembly. The seat support assembly is connected to the first front leg and the first back leg. The seat assembly may releasably connect to the seat support assembly. The seat support assembly may include a front seat support pivotally connected to the first front leg, a rear seat support pivotally connected to the first back leg, and a strut extending between the first front seat support and the first rear seat support. The seat assembly may include a pair of seat side rails and a seat fabric extending between the pair of seat side rails. The portable chair may have a use configuration and a folded configuration.



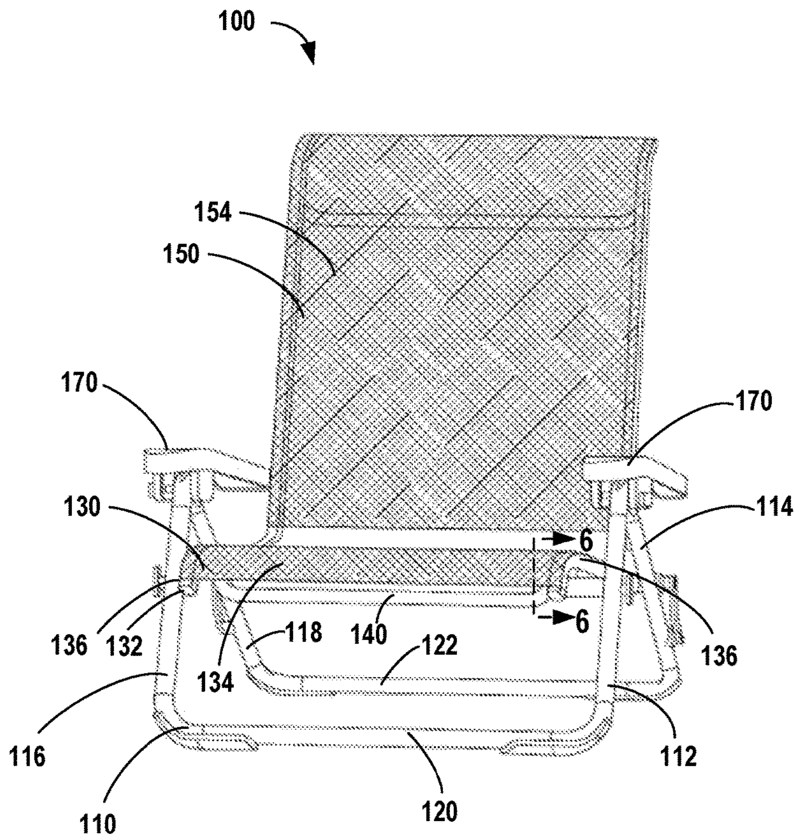

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FIG. 1

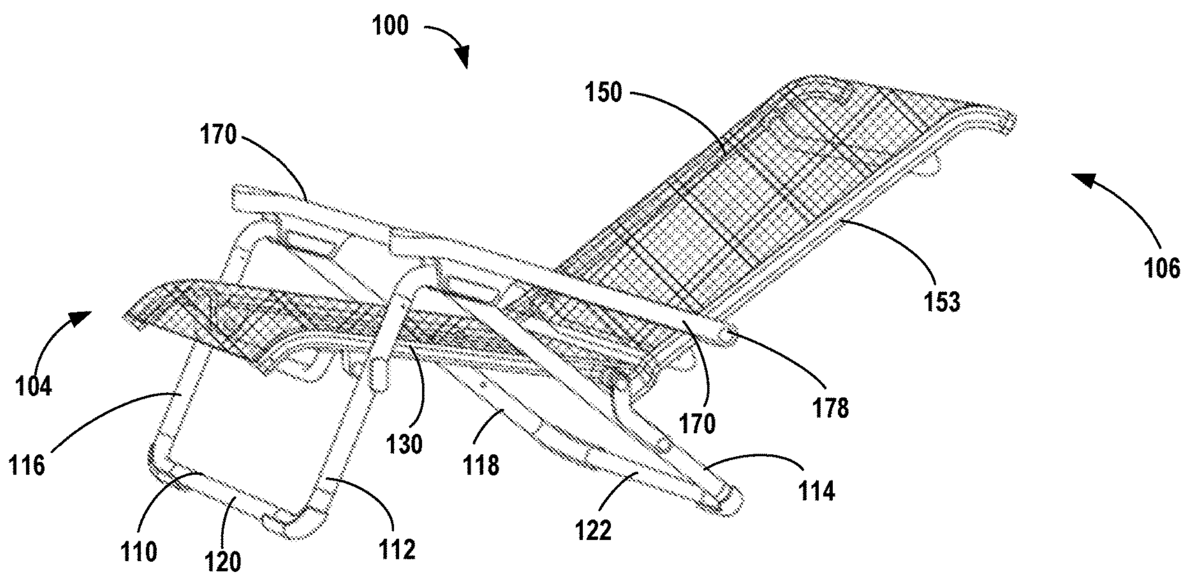
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FIG. 2

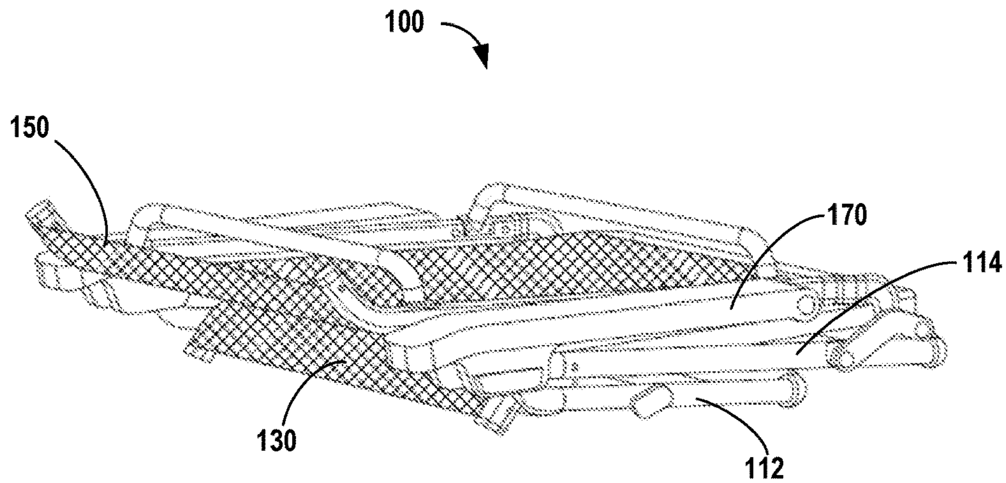

FIG. 3

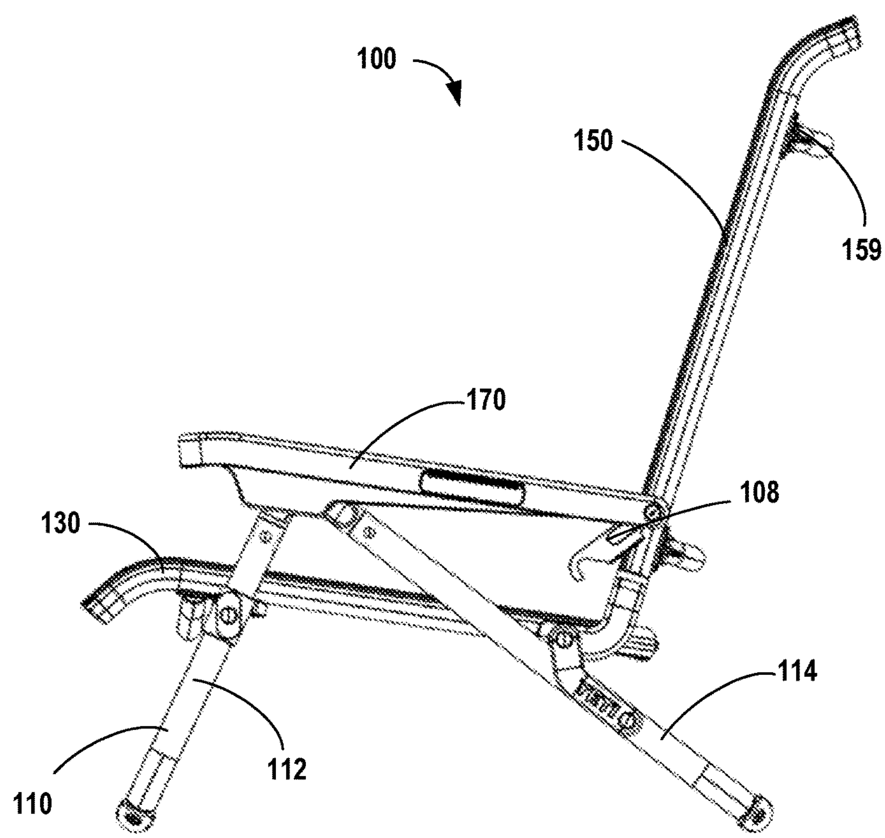

FIG. 4

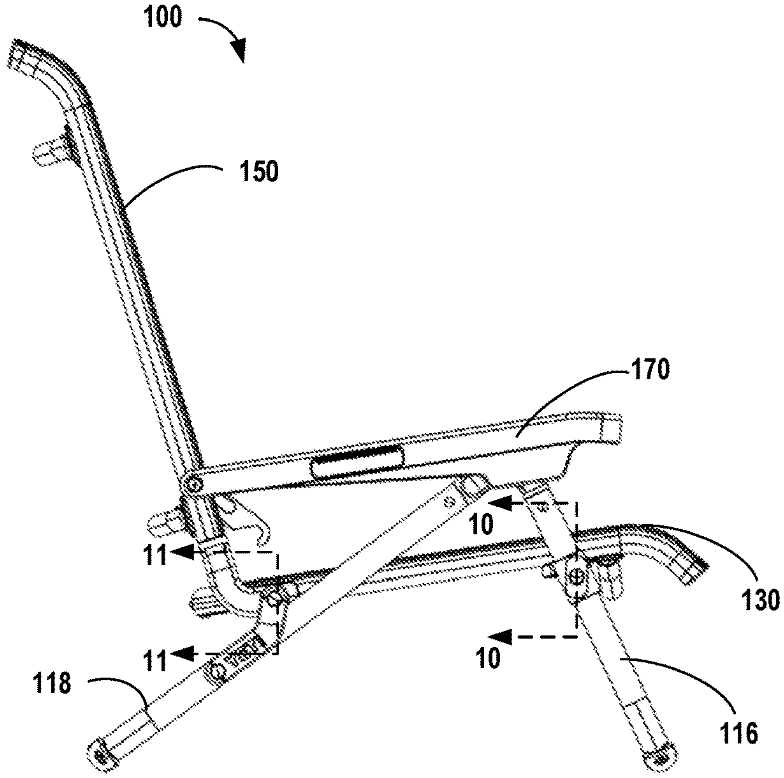

FIG. 5

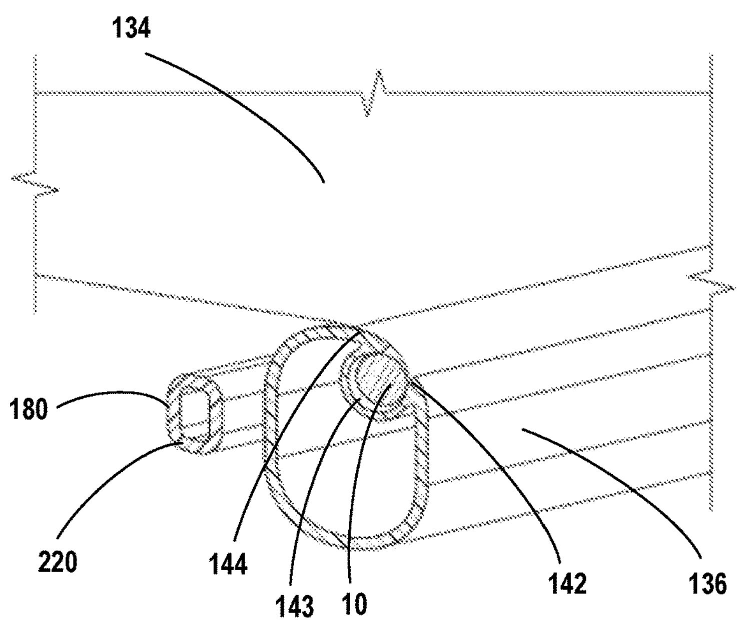

FIG. 6

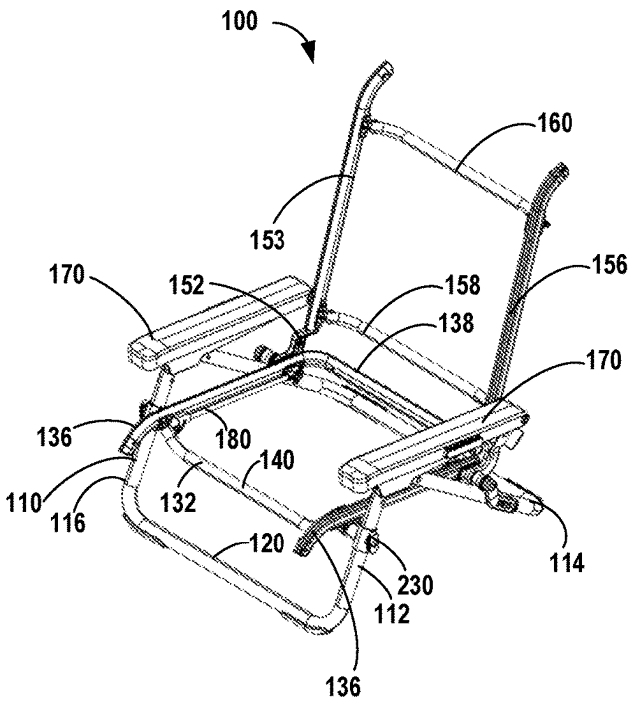

FIG. 7

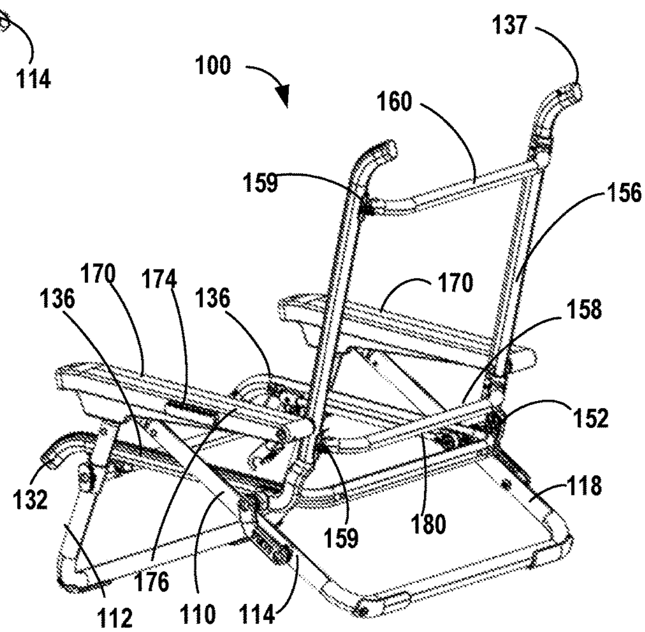

FIG. 8

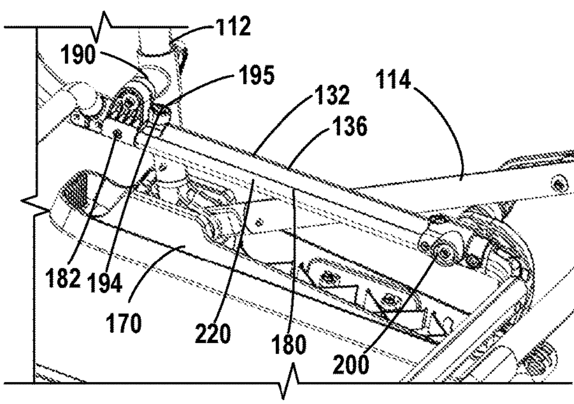

FIG. 9

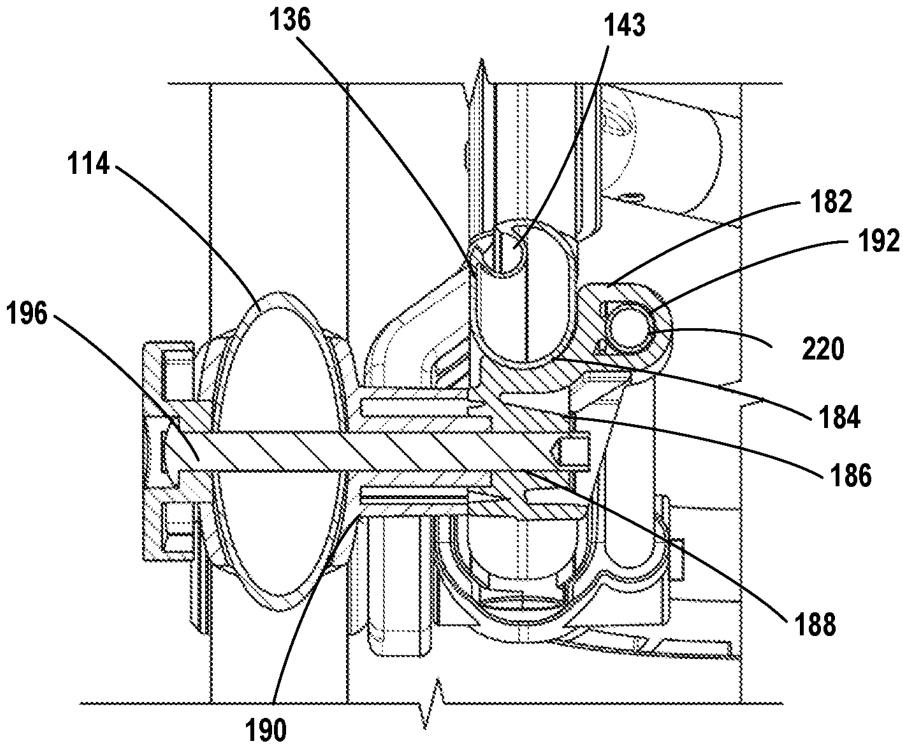
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FIG. 10

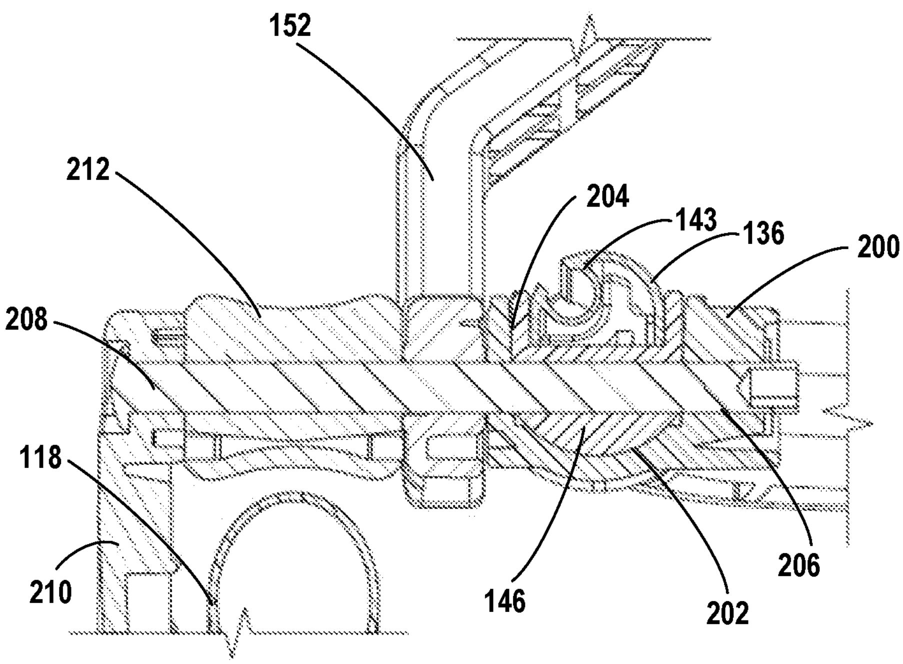
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FIG. 11

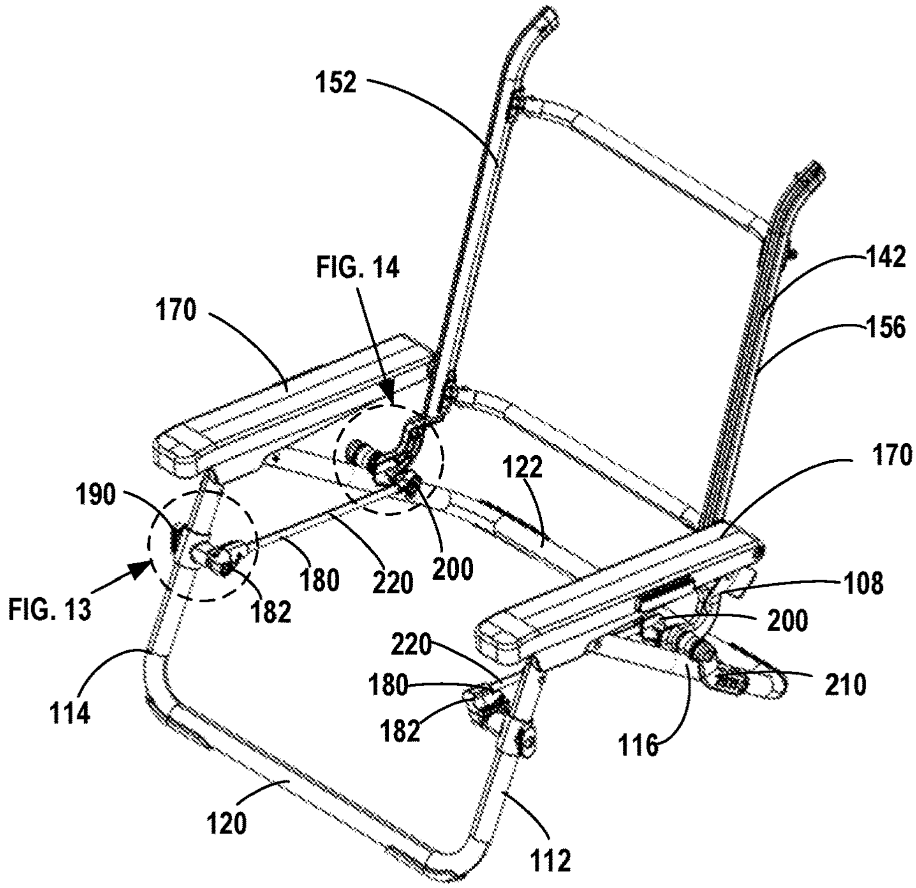
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FIG. 14
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FIG. 12

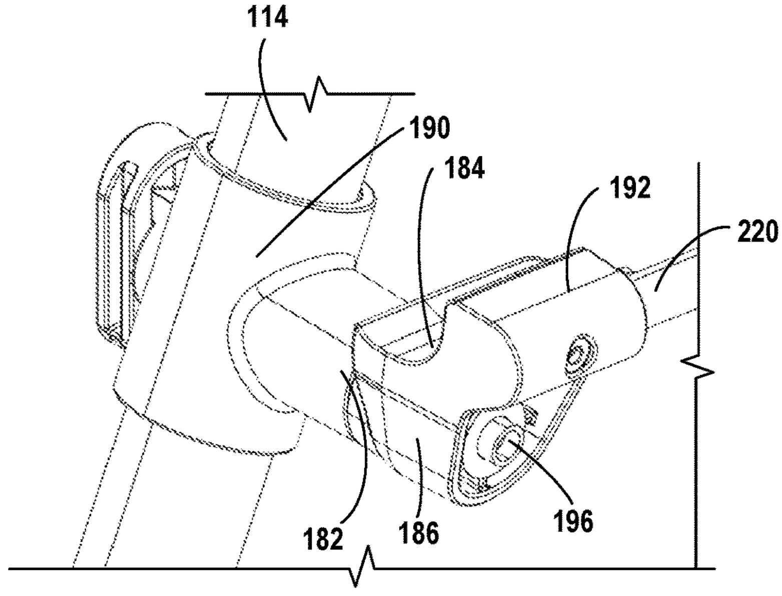
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FIG. 13

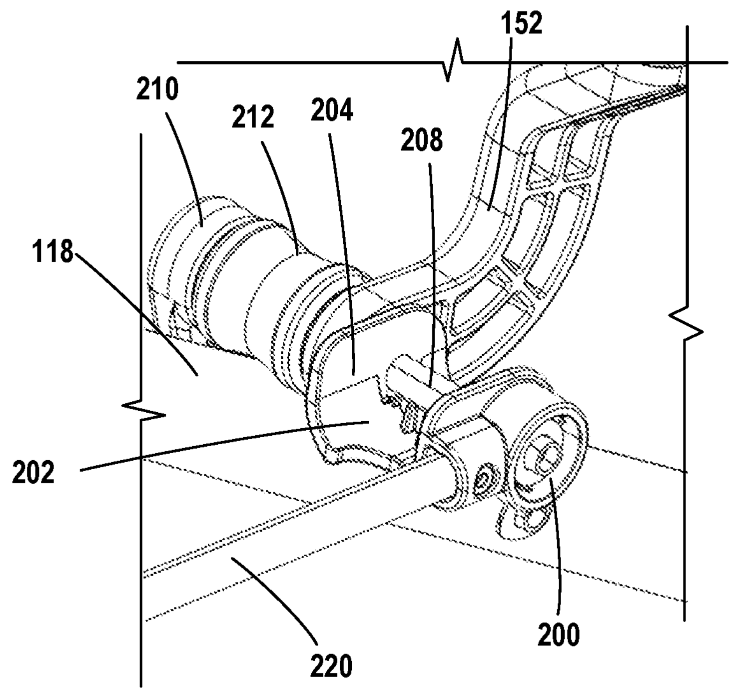

FIG. 14

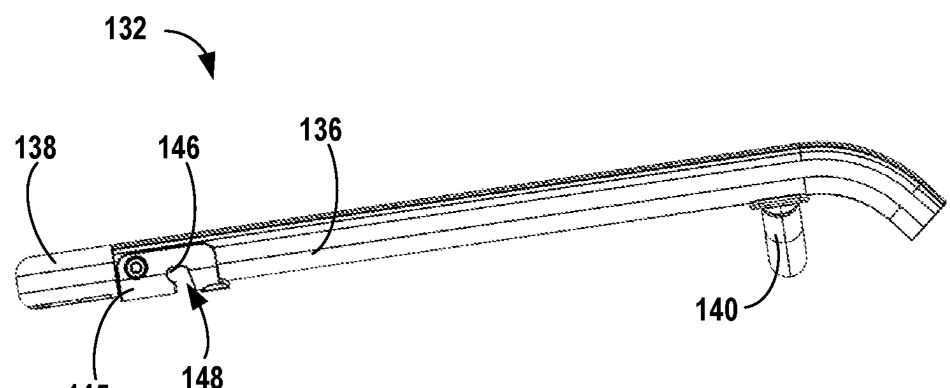

FIG. 15

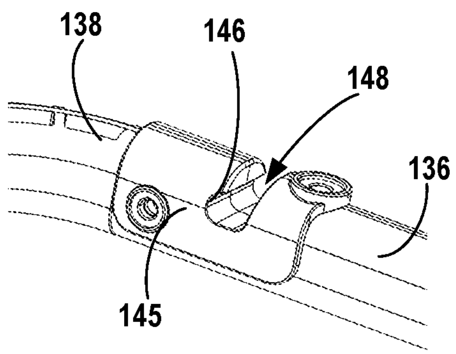

FIG. 16

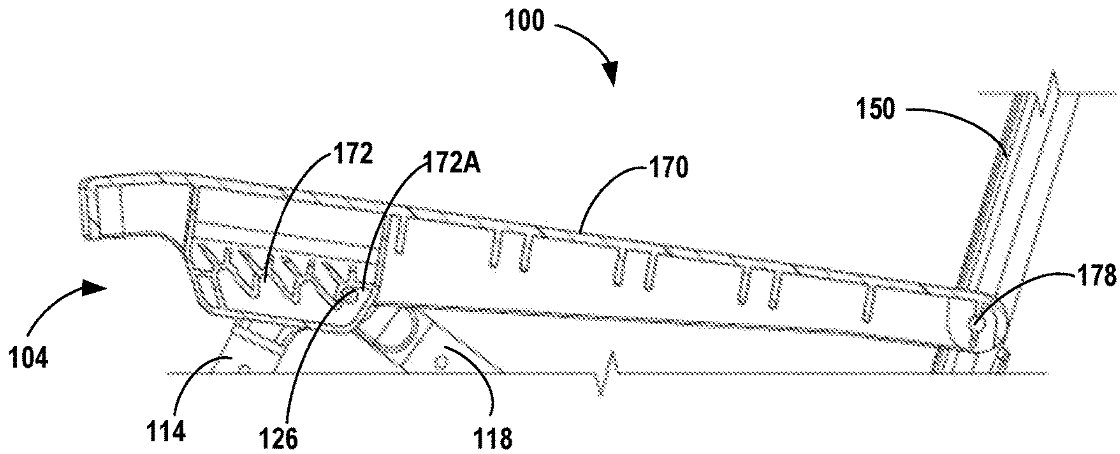
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FIG. 17

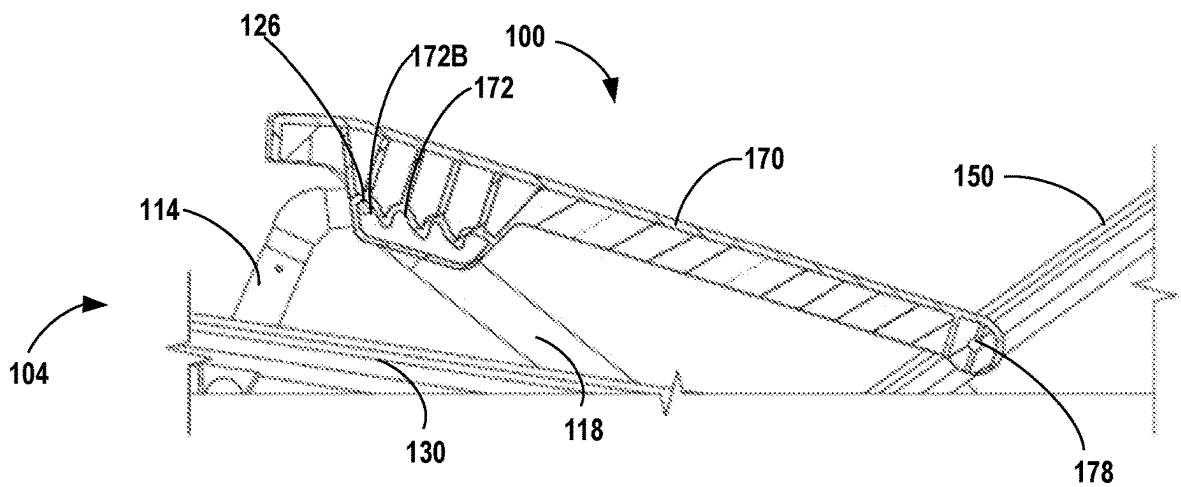
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FIG. 18

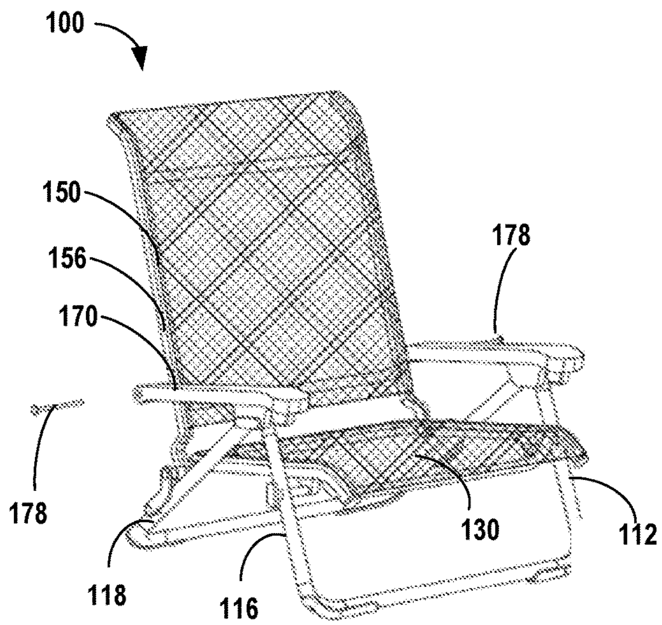

FIG. 19

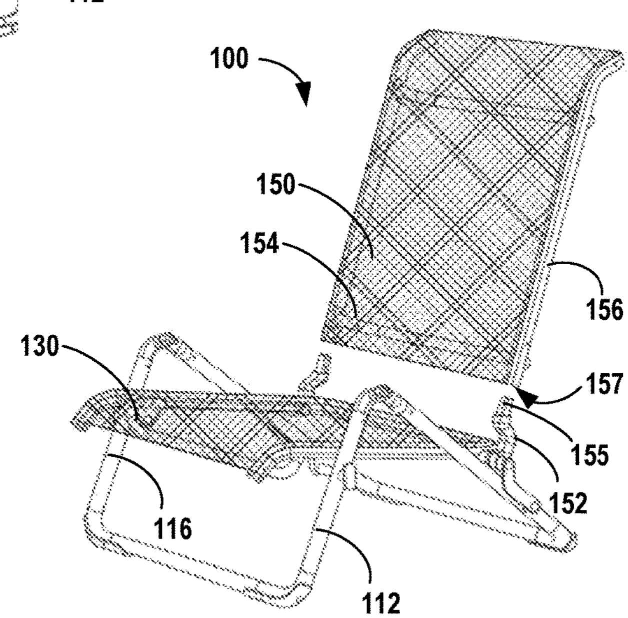

FIG. 20

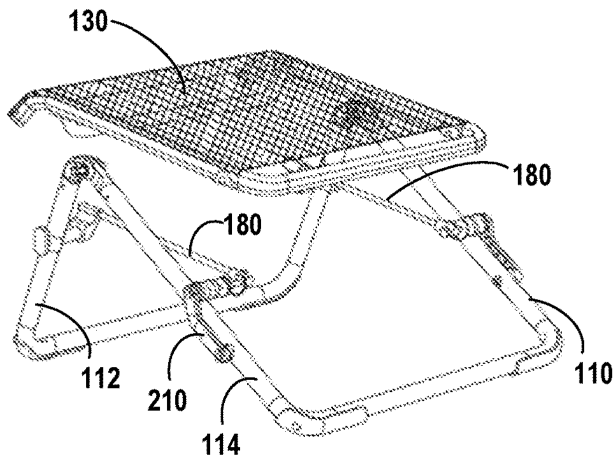

FIG. 21

PORTABLE CHAIR

FIELD OF THE INVENTION

[0001] This disclosure relates to portable chairs. More specifically, aspects of this disclosure relate to portable and collapsible chairs that are able to be adjusted for multiple seating positions.

BACKGROUND

[0002] Portable chairs are commonly used during events and activities where seating is desirable, but not always provided, such as tailgating, camping, going to the beach, and other outdoor activities. In most cases, the chairs may be uncomfortable and have limited ability to adjust the seating position. Additionally, the chairs may have poor durability reducing the long-term viability of the chair. Accordingly, overall user satisfaction with some portable chairs is low and the frequency of replacement is high.

BRIEF SUMMARY

[0003] The following presents a simplified summary of various aspects described herein. This summary is not an extensive overview, and is not intended to identify key or critical elements or to delineate the scope of the claims. The following summary merely presents some concepts in a simplified form as an introductory prelude to the more detailed description provided below.

[0004] This disclosure may relate to a portable chair that comprises: (a) a first front leg, (b) a first back leg pivotally connected to the first front leg, (c) a second front leg, (d) a second back leg pivotally connected to the second front leg, and (e) a first seat support assembly connected to the first front leg and the first back leg. The first seat support assembly may include a first front seat support pivotally connected to the first front leg, a first rear seat support pivotally connected to the first back leg, and a first strut extending between the first front seat support and the first rear seat support. A seat assembly may be releasably connected to the first seat support assembly, where the seat assembly comprises: (a) a pair of seat side rails, (b) a seat back rail extending between the pair of seat side rails, and (c) a seat fabric extending between the pair of seat side rails. A first seat side rail of the pair of seat side rails may be supported by the first front seat support and the first rear seat support. The portable chair may also have a use configuration and a folded configuration. The first strut may be located inboard of the first seat side rail. The first strut may be substantially parallel to the first seat side rail. The first front seat support may include a support surface having a concave shape that supports the first seat side rail. The first front seat support may also include an aperture below the support surface that receives a front pivot, where the front pivot pivotally connects the first front seat support and the first front leg. The first rear seat support may include a support surface that has a concave shape that supports the first seat side rail. The first rear seat support may include an aperture that extends through the support surface and receives a rear pivot, where the rear pivot may pivotally connect the first rear seat support and the first back leg. The portable chair may further include a back assembly pivotally connected to the first back leg and the second back leg, the back assembly comprising: a pair of back side rails, a back fabric extending between the pair of back side rails, a first armrest pivotally connected to a first back side rail of the pair of back side rails, and a second armrest pivotally connected to a second back side rail of the pair of back side rails. A first guide may be pivotally connected to the first front leg and the first back leg, and a second guide may be pivotally connected to the second front leg and the second back leg. The first armrest is movably connected with the first guide, and the second armrest is movably connected with the second guide. When the portable chair is in the use configuration, the portable chair may have an upright position and a reclined position. The first armrest may include a set of internal receivers that receive the first guide such that when the first guide is in a rearward most receiver of the set of internal receivers, the portable chair is in the upright position. In addition, when the first guide is in a receiver of the set of receivers located forward of the rearward most receiver, the portable chair is in a reclined position. The back assembly may be releasably connected to a pair of backrest couplers, where an upper portion of a first backrest coupler of the pair of backrest couplers is inserted into an opening of at a bottom of a first back side rail. The first back side rail may be secured to the first backrest coupler by an armrest pivot that is also pivotally connected to the first armrest.

[0005] Other aspects of this disclosure may relate to a portable chair comprising: (a) a chair frame including: (1) a first front leg, (2) a second front leg, (3) a first back leg connected to the first front leg, (4) a second back leg connected to the second front leg, (5) a first seat support assembly including: a first front seat support including a first concave surface and a first rear seat support including a first pin, and (6) a second seat support assembly including: a second front seat support including a second concave surface and a second rear seat support including a second pin, and (b) a seat assembly including: (1) a first seat side rail including a first aperture and a first convex portion, where the first aperture configured to releasably receive the first pin, and where the first concave surface is configured to releasably receive and support the first convex portion, (2) a second seat side rail including a second aperture and a second convex portion, where the second aperture configured to releasably receive the second pin, and where the second concave surface is configured to releasably receive and support the second convex portion, and (3) seat fabric extending between the first seat side rail and the second seat side rail. The chair frame may further comprise a back assembly operably connected to the first back leg and the second back leg. The back assembly may include: (a) a first back side rail, (b) a second back side rail, and (c) back fabric extending between the first back side rail and the second back side rail. A first armrest may be connected to the first back side rail, and a second armrest may be connected to the second back side rail. A first guide may pivotally connect the first front leg and the first back leg, and a second guide may pivotally connect the second front leg and the second back leg. The first armrest may be movably connected with the first guide, and the second armrest is movably connected with the second guide. A hook may be pivotally connected to the first armrest, where the hook releasably engages a portion of the chair frame to prevent the portable chair from unfolding when the portable chair is in a folded configuration. The first back side rail may be releasably connected to a first backrest coupler and the second back side rail is releasably connected to a second backrest coupler. The first backrest coupler may be operably connected to the first back leg and the second backrest coupler is operably connected to the second back leg. An upper protrusion of the first backrest coupler may be inserted into a bottom of the first back side rail. The chair frame may further comprise a first strut extending between the first front seat support and the first rear seat support, and a second strut extending between the second front seat support and the second rear seat support.

[0006] Still other aspects of this disclosure may relate to a portable chair comprising: (a) a first front leg, (b) a first back leg pivotally connected to the first front leg by a first guide, (c) a second front leg, and (d) a second back leg pivotally connected to the second front leg by a second guide. The portable chair may further comprise a back assembly including: (a) a first back side rail including a first bottom opening, (b) a second back side rail including a second bottom opening, and (c) back fabric extending between the first back side rails and the second back side rail. A first backrest coupler may be operatively coupled to the first back leg, where the first backrest coupler includes a first protrusion configured to be received in the first bottom opening of the first back side rail to removably couple the first back side rail to the first back leg, and a second backrest coupler may be operatively coupled to the second back leg, where the second backrest coupler includes a second protrusion configured to be received in the second bottom opening of the second back side rail to removably couple the second back side rail to the second back leg. A first armrest may be pivotally connected to the first back side rail with a first removable armrest pivot, where the first armrest is further connected to the first guide such that the first armrest is retained on the first guide when the first removable armrest pivot is removed, and a second armrest may also be pivotally connected to the second back side rail with a second removable armrest pivot, where the second armrest is further connected to the second guide such that the second armrest is retained on the second guide when the second removable armrest pivot is removed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** depicts a top, right front perspective view of the portable chair in a use configuration in an upright position according to aspects disclosed herein;

[0008] FIG. **2** depicts a top, right front perspective view of the portable chair of FIG. **1** in a use configuration in a reclined position according to aspects disclosed herein;

[0009] FIG. **3** depicts a perspective view of the portable chair of FIG. **1** in a folded configuration according to aspects disclosed herein;

[0010] FIG. **4** depicts a right side view of an alternate example of the portable chair of FIG. **1** in an upright position according to aspects disclosed herein;

[0011] FIG. **5** depicts a left side view of the portable chair of FIG. **4** in an upright position according to aspects disclosed herein;

[0012] FIG. **6** depicts an enlarged partial cross-sectional perspective view of the seat assembly of the portable chair of FIG. **1** along line **6-6** according to aspects disclosed herein;

[0013] FIG. **7** depicts a top, front perspective view of the portable chair of FIG. **4** with the fabric removed for clarity according to aspects disclosed herein;

[0014] FIG. **8** depicts a top, rear perspective view of the portable chair of FIG. **7** according to aspects disclosed herein;

[0015] FIG. **9** depicts a bottom, rear perspective view of the portable chair of FIG. **7** according to aspects disclosed herein;

[0016] FIG. **10** depicts an enlarged partial cross-sectional view of the portable chair of FIG. **1** along line **10-10** according to aspects disclosed herein;

[0017] FIG. **11** depicts an enlarged partial cross-sectional view of the portable chair of FIG. **1** along line **11-11** according to aspects disclosed herein;

[0018] FIG. **12** depicts a top, front perspective view of the portable chair of FIG. **1** with the seat assembly removed for clarity according to aspects disclosed herein;

[0019] FIG. **13** depicts an enlarged partial perspective view of the portable chair of FIG. **12** according to aspects disclosed herein;

[0020] FIG. **14** depicts an enlarged partial perspective view of the portable chair of FIG. **12** according to aspects disclosed herein;

[0021] FIG. **15** depicts a side view of the seat frame of the portable chair of FIG. **1** according to aspects disclosed herein;

[0022] FIG. **16** depicts a partial bottom perspective view of the seat frame of FIG. **15** according to aspects disclosed herein;

[0023] FIG. **17** depicts a side cross-sectional view through an armrest of the portable chair of FIG. **1** when the chair is in an upright position according to aspects disclosed herein;

[0024] FIG. **18** depicts a side cross-sectional view through an armrest of the portable chair of FIG. **1** when the chair is in a reclined position according to aspects disclosed herein;

[0025] FIG. **19** depicts the portable chair of FIG. **1** in a partially unassembled state according to aspects disclosed herein;

[0026] FIG. **20** depicts the portable chair of FIG. **1** in a partially unassembled state according to aspects disclosed herein; and

[0027] FIG. **21** depicts the portable chair of FIG. **1** in a partially unassembled state according to aspects disclosed herein.

DETAILED DESCRIPTION

[0028] In the following description of various example structures, reference is made to the accompanying drawings, which form a part hereof, and in which are shown by way of illustration various example devices, systems, and environments in which aspects of the disclosure may be practiced. It is to be understood that other specific arrangements of parts, example devices, systems, and environments may be utilized and structural and functional modifications may be made without departing from the scope of the present disclosure.

[0029] Also, while the terms "top," "bottom," "front," "back," "side," "rear," and the like may be used in this specification to describe various example features and elements of the invention, these terms are used herein as a matter of convenience (e.g., based on the example orientations shown in the figures or the orientation during typical use). Additionally, the term "plurality," as used herein, indicates any number greater than one, either disjunctively or conjunctively, as necessary, up to an infinite number.

[0030] The terms "connect," "couple", or "attach" as used herein indicates that components, surfaces, or features and the like may be directly or indirectly (i.e. through an intermediary) joined, linked, or attached. In addition, "pivotally connected” as used herein, indicates that the components or features may be directly or indirectly coupled together such that the components can rotate relative to each other while still being coupled together either directly or indirectly. Examples of a “pivotally connected” joint may include a pin inserted into an opening arranged in each of the components to “pivotally connect” the components.

[0031] Nothing in this specification should be construed as requiring a specific three dimensional orientation of structures in order to fall within the scope of this invention. The reader is advised that the attached drawings are not necessarily drawn to scale.

[0032] Generally, this disclosure relates to a portable chair that has an unfolded or use configuration and a folded or transport configuration. The portable chair may be easily folded and carried by a user to any location and then easily be unfolded to provide comfortable seating. The portable chair may also have multiple seating positions from an upright position to a plurality of reclined positions (i.e., in a reclined position, the angle formed between a back assembly and a seat assembly is greater than when the portable chair is in an upright position).

[0033] As shown in FIGS. **1-5**, the portable chair **100** may comprise a chair frame **110**, a seat assembly **130**, and a back assembly **150**. The seat assembly **130** and the back assembly **150** may be releasably connected to the chair frame **110**. By having a releasable seat assembly **130** and back assembly **150**, if a seat assembly **130** and/or a back assembly **150** is damaged, the seat assembly **130** and/or the back assembly **150** may be replaced with another one without having to replace the entire chair.

[0034] The chair frame **110** may include a first front leg **112**, a first back leg **114** pivotally connected to the first front leg **112**, a second front leg **116**, and a second back leg **118** pivotally connected to the second front leg **116**. The first front leg **112** may be connected to the second front leg **116** by a front sled **120**. Similarly, the first back leg **114** may be connected to the second back leg **118** by a back sled **122**. A first guide **126** may pivotally connect the first front leg **112** and the first back leg **114**, and a second guide **126** may pivotally connect the second front leg **116** and the second back leg **118**. The portable chair **100** may also include a pair of armrests **170** that are each pivotally connected to the back assembly **150** and are also each movably connected to one of the guides **126**. Each armrest **170** may be pivotally connected to the back assembly **150** by a removable armrest pivot **178** (e.g., pin). Each armrest **170** may be on opposite sides of the chair frame **110**, such that one armrest **170** is movably connected to the first guide **126** and the other armrest **170** is movably connected to the second guide **126**. In addition, the chair **100** may include a mount **230** on one or both of the front legs **112**, **116** that may be used to secure an external cupholder or other accessory (not shown).

[0035] As best shown in FIGS. **6-16**, the chair frame **110** may include a pair of seat support assemblies **180** that releasably connect to and secure the seat assembly **130** to the chair frame **110**. The seat assembly **130** may comprise a seat frame **132** and a seat fabric **134** that is connected to the seat frame **132**. The seat frame **132** may include a pair of seat side rails **136** arranged opposite each other where the seat fabric **134** extends between the pair of seat side rails **136**. Each seat support assembly **180** may be pivotally connected to either the first front leg **112** and the first back leg **114** or the second front leg **116** and the second back leg **118**. The seat assembly **130** may be releasably connected to the pair of seat support assemblies **180**. The seat support assemblies **180** may maintain the rigidity and structure of the chair frame **110** when the seat assembly **130** is removed. Thus, the seat assembly **130** may be removably connected without negatively effecting the structural integrity of the chair frame **110**.

[0036] FIGS. **7-9** illustrate the chair **100** with the fabrics **134**, **154** removed for clarity of the seat support assemblies **180**. Each seat support assembly **180** may comprise a front seat support **182**, a rear seat support **200**, and a strut **220**. Each strut **220** may extend between and connect to a front seat support **182** and a rear seat support **200**. Each seat support assembly **180** may be pivotally connected to their respective front legs **112**, **116** and the respective back legs **114**, **118**. For instance, a first seat support assembly **180** may be pivotally connected to the first front leg **112** and back leg, **114** while a second seat support assembly **180** may be pivotally connected to the second front and the back legs **114**, **118**. The front seat supports **182** may be pivotally connected to the respective front legs **112**, **116** with respective front pivots **196** (e.g., pins), and the rear seat supports **200** may be pivotally connected to the respective back legs **116**, **118** with respective rear pivots **208** (e.g., pins).

[0037] When the seat assembly **130** is connected to the pair of seat support assemblies **180**, each strut **220** of each seat support assembly **180** may be located inboard (i.e., toward a center plane that runs from a forward end **104** to a rearward end **106** of the portable chair **100**) from the corresponding seat side rail **136**. Alternatively, the seat support assembly **180** may be arranged such that the strut **220** is located outboard or below the corresponding seat side rail **136**. The seat assembly **130** may be releasably connected to each seat support assembly **180** using a mechanical fastener **195**. In some examples, the seat assembly **130** may be releasably secured to each seat support assembly **180** with only a single mechanical fastener **195**. In the illustrated example, a single fastener **195** releasably connects each seat side rail **136** of the seat frame **132** to a corresponding seat support assembly **180**. The fastener **195** may releasably connect a front seat support **182** to a seat side rail **136** of the seat frame **132**. The front seat support **182** may include an aperture **194** that receives the mechanical fastener **195** and connects to a corresponding receiver on the seat side rail **136** as shown in FIG. **9**. In the illustrated example, the receiver on the seat side rail **136** may be located on a bottom surface of the seat side rail **136**, however, the aperture **194** and corresponding receiver on seat side rail **136** may be on a top surface, side surface, or any appropriate location. Thus, by removing only two fasteners **195** (i.e., a single fastener **195** per seat side rail **136**), a user may easily replace the seat assembly **130** if desired or necessary.

[0038] As shown in FIGS. **10** and **12-13**, each front seat support **182** may include a support surface **184** that supports and contacts the respective seat side rail **136**. The support surface **184** may have a concave shape and may also have a substantially vertical portion that confronts a side surface of the respective seat side rail **136**. Each seat side rail **136** may have a convex portion. For instance, the top and bottom surfaces of each seat side rail **136** may be convex connected by substantially planar side surfaces. While in other instances, each seat side rail **136** may have a rounded, elliptical, or generally rectangular shape. The convex portion of each seat side rail **136** may be releasably received in the concave support surface **184** of the front seat support **182**.

[0039] In addition, the front seat support **182** may include a connection portion **186** that includes an aperture **188** that receives a front pivot **196** as shown in FIG. **10**. The aperture **188** may be positioned below the support surface **184**. The connection portion **186** may include an engagement member **190** that engages the respective front leg **112**, **116**. The front seat support **182** may also include a receiver **192** inboard of the support surface **184**, where the receiver **192** comprises a cavity that receives a forward end of the strut **220**.

[0040] As best shown in FIGS. **11-12** and **14**, each rear seat support **200** may include a support surface **202** that supports and contacts the respective seat side rail **136**. The support surface **202** may have a concave portion and may also have a pair of substantially vertical portions **204** that each confront side surfaces of the respective seat side rails **136**. Similar to the front seat supports **182**, a convex portion of the seat side rails **136** may be releasably received in the concave support surface **202** of the rear seat support **200**. In addition, the rear seat support **200** may include an aperture **206** that extends through the vertical portions **204** that receives a rear pivot **208**. The rear pivot **208** may connect the rear seat support **200** to a connecting member **210** that is pivotally connected to the back legs **114**, **118**. The connecting member **210** may rotate when the portable chair **100** is moved from an unfolded or use configuration to a folded configuration. In addition, a backrest coupler **152** may be pivotally connected with the rear seat support **200** and the rear pivot **208** as shown in FIGS. **11** and **14**. In addition, a roller **212** may be rotationally connected with the pivot **208** that may engage a respective back leg **114**, **118** when the portable chair **100** is moved between folded and unfolded configurations. The roller **212** may be positioned between the rear seat support **200** and the connecting member **210**.

[0041] In addition, each seat side rail **136** may releasably engage the pivot **208** with a hook **146** that is located on each seat side rail **136**. As shown in FIGS. **15-16**, the hook **146** may have an open end facing toward the forward end **104** of the chair **100**, however, the hook **146** may have an open end facing toward the rearward end **106** of the chair **100**. When the seat assembly **130** is installed onto the seat support assemblies **180**, the pivot **208** of each seat support assembly **180** may be received in an aperture **148** adjacent the hook **146**. Once the pivots **208** are received in the apertures **148**, the seat assembly **130** may be moved forward until the pivots **208** are engaging the hooks **146**. Once the hooks **146** engage the pivots **208**, the apertures **194** of the forward seat supports will align with the receivers of the seat side rails **136** to allow the mechanical fastener **195** to be inserted to releasably secure the seat assembly **130** to the chair frame **110**. The hooks **146** may be integrally formed on the seat side rails **136** or may be formed on a separate component like bracket **145** that may be fixedly connected to each of the seat side rails **136**. Alternatively, the seat support assemblies **180** and the side seat rails **136** may be arranged such that the hook **146** engages a pivot connected to a front seat support **182** and a rear seat support has a receiver to allow the mechanical fastener to engage a rear seat support **200**.

[0042] As discussed above, the seat assembly **130** may comprise a seat frame **132** and a seat fabric **134** that is connected to the seat frame **132**. The seat frame **132** may include a pair of seat side rails **136** arranged opposite each other and connected to each other by a seat back rail **138**. The seat fabric **134** may extend from each end portion across the seat frame **132** unsupported (i.e., the seat fabric **134** extends from the first seat side rail **136** to the second seat side rail **136** opposite the first seat side rail **136**). A seat rib **140** may extend between the first seat side rail **136** and the second seat side rail **136**, where the seat rib **140** has a central portion that has an upper surface that is spaced below and away from the seat fabric **134**. The seat back rail **138** may extend between ends of the pair of seat side rails **136**. The seat fabric **134** and the back fabric **154** may be configured to prevent deflection beyond a predetermined amount when supporting a user. For instance, the seat fabric **134** may be connected to the respective seat side rails **136** and then tensioned by fixedly connecting the seat rib **140** and seat back rail **138** to set the desired spacing between the seat side rails **136** while putting the seat fabric **134** in the optimal tension to support a user. Similarly, the back fabric **154** may be connected to the respective back side rails **156** and then tensioned by fixedly connecting the back ribs **158**, **160** to set the desired spacing between the back side rails **156** while putting the back fabric **154** in the optimal tension to support a user.

[0043] As shown in FIG. **6**, each seat side rail **136** and each back side rail **156** may include a slot or groove **142** that extends along a majority or even an entire length of each side rail **136**, **156**. The groove **142** may lead to a channel **143**. The slot or groove **142** may be located on an outboard portion of an upper surface **144** toward an outboard corner of each seat side rail **136** and each back side rail **156**. The seat fabric **134** may be secured to each seat side rail **136** by inserting an end portion of the seat fabric **134** into the groove **142** located on each seat side rail **136**. Similarly, the back fabric **154** may be secured to each back side rail **156** by inserting an end portion of the back fabric **154** into the groove **142**. Each end portion of the fabrics **134**, **154** may comprise a loop, where the loop is secured within the channel **143** of each side rail **136**, **156** by a rod **10** that is inserted through the loop on each end portion of the fabrics **134**, **154**. While FIG. **6** illustrates the seat fabric **134** being secured to the seat side rail **136**, the back fabric **154** may be secured to the back side rail **136** in the same manner. The width of the groove **142** may be less than the diameter of the rod **10**, thereby preventing the rod **10** from exiting the channel **143** through the groove **142** and securing the seat fabric **134** to the seat frame **132**. The seat side rails **136** and the back side rails **156** may have rounded corners, where the rails **136**, **156** have a cross-sectional shape of a rounded rectangle with flat side surfaces and rounded corners. The rounded corners help to relieve any stress on each fabric **134**, **154** as it exits the groove **142**. By relieving the stress on the fabrics **134**, **154** at the exit of the groove **142**, the durability and life of the fabrics **134**, **154** may be greatly increased. Each side rail **136**, **156** may include an end cap **137** that helps to further secure each fabric **134**, **154** to its corresponding side rail **136**, **156**.

[0044] Additionally, chair **100** may have a symmetric construction where components on a left side of the chair **100** may be mirrored to the components of the right side of the chair **100** (i.e., the front leg **112**, **116**, the back legs **114**, **118**, the seat support assemblies **180**, the seat side rails **136**, the backrest couplers **152**, and the back side rails **156** may be mirror images of each when mirrored through a center plane that extends from the forward end **104** to the rearward end **106**).

[0045] The back assembly **150** may include a removable back frame **153** that supports a back fabric **154**. The back frame **153** may be removably connected to the backrest couplers **152**. The back frame **153** may include a pair of back side rails **156** arranged opposite each other, a lower back rib **158** extending between the back side rails **156**, and an upper back rib **160** extending between the back side rails **156** and spaced apart from and above the lower back rib **158**. The back ribs **158**, **160** may be substantially parallel to each other. In addition, the back ribs **158**, **160** provide structural stability to the back frame **153**. The back ribs **158**, **160** may be spaced away from and behind the back fabric **154**. In addition, the back side rails **156** may include one or more grommets **159** extending rearward from an outer surface. Each back side rail **156** may include a grommet **159** in an upper portion and a grommet **159** in a lower portion such that the grommets **159** on each back side rail **156** are spaced apart from each other. The grommets **159** may be configured to receive a strap or rope (not shown) or other apparatus to assist in carrying the chair **100** or attaching the chair **100** to another item. In some examples, the grommets **159** may be arranged only on one of the back side rails **156**.

[0046] The portable chair **100** may be easily converted from the use configuration shown in FIG. **1** to a folded configuration shown in FIG. **3** for easy transport to another location or for storage. Through actuation of the rotational and pivotable interfaces, the portable chair **100** may be folded into a folded configuration as shown in FIG. **3** and unfolded into a use configuration as shown in FIGS. **1** and **2**. In the folded configuration, the front legs **112**, **116** may be substantially parallel to back legs **114**, **118** and seat assembly **130** may be substantially parallel to back assembly **150**. While in the folded configuration, the chair **100** may be locked and/or sustained in the folded configuration by a hook **108**. The hook **108** may be pivotally connected to an armrest **170** and in some examples, may be pivotally connected to the back assembly **150**. The hook **108** may releasably engage the connecting member **210** to prevent the chair **100** from unintentionally unfolding. In some examples, the hook **108** may engage other components of the chair frame **110** such as the back legs **114**, **118**, the rear seat support **200**, or a portion of the chair frame **110**.

[0047] The two front legs **112**, **116**, the two back legs **114**, **118**, the front sled **120**, the back sled **122** along with the rails and ribs **136**, **138**, **140**, **156**, **158**, **160** of the seat frame **132** and the back frame **153** may be cylindrical tubes, rectangular tubes with rounded corners, and/or shafts, or other hollow shape. The front legs **112**, **116** and front sled **120** may be formed as an integral, single part. Similarly, the back legs **114**, **118** and back sled **122** may be formed as an integral, single part. These components may be made of, for example, aluminum, titanium, stainless steel, scandium, metal alloys, polymers, composites, carbon fiber, and/or wood, such as bamboo. In instances in which aluminum, titanium, stainless steel, scandium, and/or metal alloys are used in the fabrication of the two front legs **112**, **116**, the two back legs **114**, **118**, the front sled **120**, the back sled **122** along with the rails **136**, **156**, **160** of the seat frame **132** and the back frame **153**, the metallic components may be hydroformed, cast, extruded, or formed by another method known to one skilled in the art. Furthermore, the metallic components may be treated through anodizing, plating, painting, powder coating, and/or the application of enamel in order to prevent corrosion induced by environmental conditions such as salt spray. Additionally, the metals and alloys used in the fabrication of the legs **112**, **114**, **116**, **118**, the sleds **120**, **122**, and the rails and ribs **136**, **138**, **140**, **156**, **158**, **160** of the frames **132**, **153** may be treated through annealing, case hardening, precipitation strengthening, tempering, normalizing, and/or quenching in order to increase hardness, toughness, and tensile and shear strength.

[0048] The fabrics **134**, **154** may be a weave-type and/or mesh-like fabric. Additionally, the fabrics **134**, **154** may be composed of any of a number of materials including, but not limited to, armored fabric cloth, sail fabric, awning fabric, Kevlar, tarp canvas, vinyl coated polyester, nylon mesh, neoprene, aluminized nylon, and/or cotton canvas. In some embodiments, the material may be treated to provide increased UV stabilization and weathering resistance, fire resistance, abrasion and tear resistance, and waterproofing. Additionally, the fabrics **134**, **154** may be constructed of a weaved material with yarn having elastomeric properties. The elastomeric properties include the ability to stretch and deform under stress (i.e., increased elasticity), such as tension or weight. The elastomeric properties allow the fabric to return to its original form and the ability to resist creep and/or permanent deformation when the stress from the load is removed. In one example, the fabrics **134**, **154** can be formed as a first yarn formed of an acrylic or polymer and blends and a second yarn formed of an elastomeric material such that the second yarn is more elastomeric than the first yarn. The elastomeric properties of the second yarn can help to provide the elastomeric properties of the fabrics **134**, **154**. In some instances, the fabrics **134**, **154** may be composed of a similar materials such that the fabric **134** of the seat assembly **130** is the same as the fabric **154** of the back assembly **150**. However, in some cases, the fabric **134** of the seat assembly **130** may be a different material than fabric **154** of the back assembly **150**. For example, seat fabric **134** may be made of a first material and/or combination of materials, and back fabric **154** may be made of a second material and/or combination of materials different than the first material and/or combination of materials.

[0049] In addition, the portable chair **100** may be arranged in multiple seating positions from an upright position to multiple reclined positions. As shown in FIGS. **1** and **2**, the portable chair **100** may be adjusted from an upright position shown in FIG. **1** to the reclined position illustrated in FIG. **2**. In the upright position, the seat assembly **130** and the back assembly **150** may form an angle of approximately 99 degrees, while in the fully reclined position the seat assembly **130** and the back assembly **150** may form an angle of approximately 136 degrees. The chair **100** may have multiple reclined positions between the upright position and the fully reclined position. Accordingly, the chair **100** may have angular difference of approximately 37 degrees between the upright position and the fully reclined position, and may have an angular difference within a range of 30 degrees and 40 degrees between the upright position and the fully reclined position.

[0050] The adjustment of the portable chair **100** may be controlled by the relative position of each guide **126** relative to its corresponding armrest **170**. As discussed above, each guide **126** is movably connected with a corresponding armrest **170** as well as pivotally connected to one of the first front leg **112** and the first back leg **114** or pivotally connected to one of the second front leg **116** and the second back leg **118**. As shown in the cross-sectional views illustrated in FIGS. **17** and **18**, each armrest **170** may include a set of internal receivers **172** that individually receive the guide **126**. The position (i.e., the angle) of the back assembly **150** relative to the seat assembly **130** depends upon which receiver **172** the guide **126** is located. In some examples, each armrest **170** may have two sets of internal receivers **172** arranged on opposite sides of each armrest **170**. Thus, the guide **126** may engage a receiver **172** in both sets of receivers **172** on both sides of each armrest concurrently. FIG. **17** illustrates the portable chair **100** is in an upright position, where the guide **126** is located in the rearward most receiver **172**A. With the guide **126** in the rearward most receiver **172**A, the armrest **170** is moved forward relative to the upper portions of the legs **112**, **114**, **116**, **118** and the back assembly **150** is rotated forward. As the guide **126** is moved to each subsequent forward receiver **172** compared to the rearward most receiver **172**A, the armrest **170** moves rearward relative to the upper portions of the legs **112**, **114**, **116**, **118** and the back assembly **150** rotates rearward to a reclined position. FIG. **18** illustrates the location of the guide **126** is in the forwardmost receiver **172**B, which corresponds with the chair **100** being in the fully reclined position. Each receiver **172** may have an opening at a bottom end of the receiver **172** with tapered sidewalls where each sidewall is angled toward a rearward end **106** of the chair **100**. The tapered sidewalls of each receiver **172** help to direct the guide **126** into the selected receiver **172**. The armrests **170** may also have an elongated hanger **174** located on an outboard surface **176** of each armrest **170**. The hanger **174** may have an elongated shape and comprise a substantially U-shaped groove. In some examples, the hanger **174** may have a retention member that extends into an upper portion of the groove to help retain the items located within the groove. The hanger **174** may be configured to receive a strap or band to support an external bag or other item (not shown). The hanger **174** may be integral with the armrest **170** such that the hanger **174** is formed with the armrest **170**, or the hanger **174** may be releasably connected (e.g., to the armrest **170** as a separate component).

[0051] The chair **100** may be easily adjusted to a desired recline position. In order to adjust the chair **100** from an upright position to a reclined position, a user may lean forward to relieve any pressure on the back assembly **150**, then lift the armrests **170** and move the armrests **170** backward (i.e., away from the forward end **104** of the chair **100**) such that the guide **126** engages a more forward receiver **172** than the previous receiver **172**. A similar process may be used to move the chair **100** from a reclined position to a more upright position. To move the chair **100** from a reclined position to a more upright position, a user may lean forward to relieve any pressure on the back assembly **150**, then lift the armrests **170** and move the armrests **170** or the back assembly **150** forward (i.e., toward from the forward end **104** of the chair **100**) such that the guide **126** engages a more rearward receiver **172** than the previous receiver **172**. In some examples, to move the chair **100** from a reclined position to a more upright position, a user may simply relieve pressure on the back assembly **150** and pull the back assembly **150** forward such that the guide **126** and the receivers **172** act in a ratcheting arrangement such that the guide **126** is received in a more rearward receiver **172** than the previous receiver **172**. As shown in FIG. **17**, the chair **100** may be in its most upright position when the guide **126** is located in the rearward most receiver **172**A. When the chair **100** is in the folded configuration, the guides **126** may be disengaged from the receivers **172**.

[0052] FIGS. **19-20** illustrate the removal of the back assembly **150** from the chair **100**. As discussed above, the back assembly **150** may be replaceable by a user. For example, the back assembly **150** may be replaced if the back assembly **150** is damaged or if a user wants a different back fabric **154**. As discussed above, the back assembly **150** may be releasably connected to the chair **100**. The back frame **153** may be releasably connected to the backrest couplers **152**. Each back side rail **156** may have a bottom opening **157** to allow for an upper protrusion **155** of each backrest coupler **152** to be inserted. Alternatively, each back side rail **156** may include a protrusion that is received in an upper opening of each backrest coupler **152** to releasably connect the back frame **153** to the backrest couplers **152**. An armrest pivot **178** may be inserted into an aperture in each armrest **170** and then through an aperture in the back side rail **156** to secure both the armrest **170** and the backrest coupler **152** to the back side rail **156**. In some examples, the armrest pivot **178** may be secured via a threaded connection. The threaded connection may be located in the armrest **170**, the backrest coupler **152**, the back side rail **156**, or a separate mechanical element that engages the armrest pivot **178**. To replace the back assembly **150**, the armrest pivots **178** may be removed to disengage each armrest **170** from its corresponding back side rail **156**. Each armrest **170** remains connected to the chair frame **110** by each guide **126** that are coupled to their respective armrest **170**. Additionally, once the armrest pivots **178** are removed, each back side rail **156** becomes slidably engaged with its corresponding backrest coupler **152**. The back assembly **150** may then be slid off the backrest couplers **152** to allow it to be replaced as shown in FIG. **20**. Thus, the back assembly **150** may be removed by removing the two armrest pivots **178** and sliding the back assembly **150** upward from the backrest couplers **152**. FIG. **21** illustrates the seat assembly **130** being removed from the chair **100**. As discussed above, the seat assembly **130** may be removed from the seat support assemblies **180** by removing the mechanical fasteners **195** and then lifting the seat frame **132** from the seat support assemblies **180**. The seat support assemblies **180** help maintain the structural integrity of the chair frame **110** with the seat assembly **130** removed. While FIG. **21** also shows the back assembly **150** removed, the seat assembly **130** may be removed without removing the back assembly **150**.

[0053] While various embodiments have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the claims. The various dimensions described above are merely exemplary and may be changed as necessary. Accordingly, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the claims. Therefore, the embodiments described are only provided to aid in understanding the claims and do not limit the scope of the claims.

We claim:

1. A portable chair comprising:

a first front leg;

a first back leg pivotally connected to the first front leg;

a second front leg;
a second back leg pivotally connected to the second front leg;
a first seat support assembly connected to the first front leg and the first back leg,
wherein the first seat support assembly comprises:
a first front seat support pivotally connected to the first front leg;
a first rear seat support pivotally connected to the first back leg; and
a first strut extending between the first front seat support and the first rear seat support;
a seat assembly releasably connected to the first seat support assembly, wherein the seat assembly comprises:
a pair of seat side rails;
a seat back rail extending between the pair of seat side rails; and
a seat fabric extending between the pair of seat side rails; and
wherein a first seat side rail of the pair of seat side rails is supported by the first front seat support and the first rear seat support; and
wherein the portable chair has a use configuration and a folded configuration.

2. The portable chair of claim **1**, wherein the first strut is located inboard of the first seat side rail.

3. The portable chair of claim **1**, wherein the first strut is substantially parallel to the first seat side rail.

4. The portable chair of claim **1**, wherein the first front seat support includes a support surface having a concave shape that supports the first seat side rail.

5. The portable chair of claim **4**, wherein the first front seat support includes an aperture below the support surface that receives a front pivot, wherein the front pivot pivotally connects the first front seat support and the first front leg.

6. The portable chair of claim **1**, wherein the first rear seat support includes a support surface that has a concave shape that supports the first seat side rail.

7. The portable chair of claim **6**, wherein the first rear seat support includes an aperture that extends through the support surface and receives a rear pivot, wherein the rear pivot pivotally connects the first rear seat support and the first back leg.

8. The portable chair of claim **1**, further comprising:
a back assembly pivotally connected to the first back leg and the second back leg, the back assembly comprising:
a pair of back side rails;
a first back side rail;
a second back side rail; and
back fabric extending between the first back side rail and the second back side rail;
a first armrest pivotally connected to a first back side rail of the pair of back side rails; and
a second armrest pivotally connected to a second back side rail of the pair of back side rails.

9. The portable chair of claim **8**, wherein a first guide pivotally connects the first front leg and the first back leg, and a second guide pivotally connects the second front leg and the second back leg; and
wherein the first armrest is configured to releasably engage the first guide, and the second armrest is configured to releasably engage the second guide.

10. The portable chair of claim **9**, wherein when the portable chair is in the use configuration, the portable chair has an upright position and a reclined position, and
wherein the first armrest includes a set of internal receivers that receive the first guide such that when the first guide is received in a rearward most receiver of the set of internal receivers, the portable chair is in the upright position and when the first guide is received in a receiver of the set of internal receivers located forward of the rearward most receiver, the portable chair is in the reclined position.

11. The portable chair of claim **8**, wherein the back assembly is releasably connected to a pair of backrest couplers, and wherein an upper portion of a first backrest coupler of the pair of backrest couplers is inserted into a bottom opening of the first back side rail.

12. The portable chair of claim **11**, wherein the first back side rail is secured to the first backrest coupler by an armrest pivot that is also pivotally connected to the first armrest.

13. A portable chair comprising:
a chair frame including:
a first front leg;
a second front leg;
a first back leg connected to the first front leg;
a second back leg connected to the second front leg;
a first seat support assembly including:
a first front seat support including a first concave surface; and
a first rear seat support including a first pin;
a second seat support assembly including:
a second front seat support including a second concave surface; and
a second rear seat support including a second pin; and
a seat assembly including:
a first seat side rail including a first aperture and a first convex portion, wherein the first aperture configured to releasably receive the first pin, and wherein the first concave surface is configured to releasably receive and support the first convex portion;
a second seat side rail including a second aperture and a second convex portion, wherein the second aperture configured to releasably receive the second pin, and wherein the second concave surface is configured to releasably receive and support the second convex portion; and
seat fabric extending between the first seat side rail and the second seat side rail.

14. The portable chair of claim **13**, wherein the chair frame further comprises:
a back assembly operably connected to the first back leg and the second back leg, the back assembly comprising:
a first back side rail;
a second back side rail;
back fabric extending between the first back side rail and the second back side rail; and
a first armrest connected to the first back side rail; and
a second armrest connected to the second back side rail.

15. The portable chair of claim **14**, wherein a first guide pivotally connects the first front leg and the first back leg, and a second guide pivotally connects the second front leg and the second back leg; and wherein the first armrest is movably connected with the first guide, and the second armrest is movably connected with the second guide.

16. The portable chair of claim **14**, wherein a hook is pivotally connected to the first armrest, wherein the hook releasably engages a portion of the chair frame to prevent the portable chair from unfolding when the portable chair is in a folded configuration.

17. The portable chair of claim **14**, wherein the first back side rail is releasably connected to a first backrest coupler and the second back side rail is releasably connected to a second backrest coupler; and

wherein the first backrest coupler is operably connected to the first back leg and the second backrest coupler is operably connected to the second back leg.

18. The portable chair of claim **17**, wherein an upper portion of the first backrest coupler is inserted into an opening at a bottom of the first back side rail.

19. The portable chair of claim **13**, wherein the chair frame further comprises:

a first strut extending between the first front seat support and the first rear seat support, and a second strut extending between the second front seat support and the second rear seat support.

20. A portable chair comprising:

a first front leg;

a first back leg pivotally connected to the first front leg by a first guide;

a second front leg;

a second back leg pivotally connected to the second front leg by a second guide;

a back assembly including:

a first back side rail including a first bottom opening;

a second back side rail including a second bottom opening; and

back fabric extending between the first back side rails and the second back side rail;

a first backrest coupler operatively coupled to the first back leg, wherein the first backrest coupler includes a first protrusion configured to be received in the first bottom opening of the first back side rail to removably couple the first back side rail to the first back leg;

a second backrest coupler operatively coupled to the second back leg, wherein the second backrest coupler includes a second protrusion configured to be received in the second bottom opening of the second back side rail to removably couple the second back side rail to the second back leg;

a first armrest, wherein the first armrest is pivotally connected to the first back side rail with a first removable armrest pivot, and wherein the first armrest is further connected to the first guide such that the first armrest is retained on the first guide when the first removable armrest pivot is removed; and

a second armrest, wherein the second armrest is pivotally connected to the second back side rail with a second removable armrest pivot, and wherein the second armrest is further connected to the second guide such that the second armrest is retained on the second guide when the second removable armrest pivot is removed.

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